

Employee Performance in Response to Workplace Errors: Evidence from the Field*

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ABSTRACT: This paper studies how holding employees accountable for their errors affects their future performance. While errors are ubiquitous, the behavioral consequences of being confronted with past errors by their managers remain unclear. Using proprietary performance and error records from more than 3,200 service technicians across 50 firms, this paper shows that such confrontations do not improve output quality and reduce customer satisfaction. Instead, output quantity increases by 5-7%, suggesting that employees are reallocating effort towards tasks that are more readily observable to their managers. These effects are driven by employees with shorter tenure and lower prior performance who have stronger incentives to restore their reputation. A back-of-the-envelope calculation suggests that the positive effect on output quantity is twice as large as the negative effect on customer satisfaction. These findings contribute to the field literature on multitasking and the broader literature on reputational concerns in firms.

Keywords: Errors, Performance, Multitasking, Reputation

JEL Codes: J24, M53

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1 Introduction

Errors are an inevitable part of work and can impose substantial costs on firms and their customers. Firms therefore specify a set of desired actions, monitor compliance, and hold employees responsible for deviations (Merchant & Van der Stede, 2017). These processes are common across business functions – from internal audits to code review in software development and quality control in manufacturing. Prior research on these control mechanisms has shown how firms can improve the detection and reporting of errors and misconduct (Altamuro & Beatty, 2010; Dyck et al., 2010), and employees’ willingness to disclose errors on their own (e.g. Gold et al., 2014). While these studies help to understand how firms prevent and uncover errors, less is known about the behavioral consequences of holding employees accountable for their errors.

This paper studies how confronting employees with past mistakes affects future performance. Errors, defined as unintentional, but avoidable deviations from the work standard (Frese & Keith, 2015), carry the implication that the employee was either *not capable* or *not careful* enough to perform their job correctly. This negative impression can have a substantial impact on the employee’s career, which may lead employees to work harder in order to “make up for the mistake” (Holmström, 1999). Yet, performance measures differ in how clearly they reveal an improvement to the supervisor. If errors are rarely observed, avoiding another one may provide little immediate evidence of improved performance. Increasing performance in a more routinely recorded performance measure might offer a more visible alternative to demonstrate competence (Brüggen, 2011; Moers, 2006). Employees may therefore shift effort towards more readily observable dimensions to put themselves in a better light as quickly as possible – potentially at the expense of other dimensions of the job (Feltham & Xie, 1994; Holmström & Milgrom, 1991; Manthei & Sliwka, 2019). If however, conversations about errors deliver knowledge that is useful across tasks (e.g. Eyring, 2020), performance could improve more broadly, reducing the need to re-allocate effort from one task to another.

To study these dynamics, this paper combines individual performance data from 3,265 employees in 50 firms with 1,161 detailed error reports, which yields around 167,000 employee-week observations. All firms are service providers for a large technology company (hereinafter referred to as *MultiCo*), whose employees work on the same infrastructure as the service providers. Employees complete around 4-5 work assignments per day, in which they install and maintain products and services from MultiCo. MultiCo's own technicians work on the same infrastructure and can report errors made by subcontractor employees through a dedicated system. Reports describe the error and its possible cause and are reviewed by MultiCo's quality management and the subcontractor. If a report is considered justified, the subcontractor's manager must contact the responsible employee and document the response. These records make it possible to link the accountability process to subsequent output quantity and quality.

Using an event-study approach, this paper shows how employee performance changes after the manager has confronted an employee with an error for the first time. Error discovery depends in part on the timing of the visits from MultiCo technicians, which creates natural variation in when errors are reported. Results show that output quantity, as measured by the average number of completed work assignments per day, sharply increases around the time point of the manager's intervention. Contrary to that, output quality does not improve – if anything, employees provide worse customer service. The results thus indicate an increase in output quantity alongside a deterioration in customer service. An illustrative cost-benefit calculation values the additional output at roughly twice the estimated cost of lower customer satisfaction.

Additional analyses using propensity score matching indicate that employees who experience an intervention are more likely to receive another error report within three or six months than otherwise similar employees without a previous report. Together with the decline in customer satisfaction, this pattern provides little support for broad learning as the main explanation for the performance increase. Further heterogeneity results are consistent with reputation repair

as a motive for this response. The increase in output is larger for employees with lower prior performance and shorter tenure, whose standing with the supervisor may be less secure or less established.

This paper makes three contributions. First, it extends research on errors in firms by examining employee behavior after an error has been detected and addressed by a manager. Laboratory research shows that career concerns can distort effort allocation (Brüggen, 2011), and field evidence links subjective evaluation to employees' efforts to influence their evaluators (De Janvry et al., 2023). I examine responses to a specific unfavorable performance signal in an ongoing employment relationship, where supervisors' judgments have real career consequences. The stronger responses among employees with lower prior performance and shorter tenure are consistent with incentives to restore a damaged reputation. Second, the paper adds to field research on multitasking and management controls. Prior field experiments show that access to performance information and changes in incentives affect effort allocation and attention (Hong et al., 2018; Kohler et al., 2024; Manthei & Sliwka, 2019). The results show a similar pattern following a managerial response to an individual error: output increases in a readily observable dimension while customer service is reduced. This evidence suggests that accountability interventions can change employees' priorities across tasks. Finally, the results contribute to a broader literature on behavioral responses to organizational policies, such as employee responses to termination notices or wage cuts (Alfitian & Vogelsang, 2026; Cederlöf et al., 2024; Krueger & Friebe, 2022; Sandvik et al., 2021). In contrast to these findings, the results indicate that seemingly "bad news" from an employee perspective can lead to a persistent increase in output – although this increase comes at the expense of a less visible dimension of performance.

The paper proceeds as follows: Section 2 lays out the guiding hypotheses of this paper. In Section 3, I describe the organizational context and the error management process which constitutes the basis for the analysis, as well as the data and identification strategy for the main

analyses. Section 4 and Section 5 show the corresponding results as well as the exploration of potential mechanisms for the main results. Section 6 presents an illustrative cost-benefit analysis, and Section 7 concludes.

2 Theory

2.1 Errors as a signal of employee ability

Frese and Keith (2015) define errors as unintentional – yet avoidable – deviations from specific goals that arise from insufficient knowledge, inadequate judgement, or flawed execution. An error is therefore different from misconduct which requires an explicit malicious intent for personal gain. This definition is similar to existing definitions for related types of errors, such as routine errors (Maletta & Wright, 1996) or mechanical errors (Owhoso et al., 2002; Ramsay, 1994). In line with this definition, errors imply that the responsible person was either incapable or not careful enough to perform the task correctly. Such failures can threaten employees' own self-concept, as they conflict with the basic need for competence (Baumeister & Tice, 1985; Deci & Ryan, 1985).¹ Importantly, however, in an organizational setting, errors are usually observed by others, most notably: supervisors. In many cases, supervisors cannot perfectly observe an employee's underlying effort or ability and have to rely on imperfect, but observable, outcomes. Errors provide one such signal and may lead supervisors to revise their assessment of the employee's capabilities downward. These assessments matter because supervisors use them for performance evaluations, task assignments and other employment decisions. Grabner & Moers (2013), for instance, show that supervisors use both current performance and subjective assessments of employee ability when making promotion decisions. Similarly, Künneke (2024) shows that supervisors differ systematically in how they evaluate employees, and that these evaluation styles have persistent consequences for employees' promotion prospects and

¹A similar mechanism has been documented in literature on negative performance feedback, which can direct attention away from the task and toward the self, thereby threatening recipients' self-concept or self-esteem (for reviews see Kluger & DeNisi (1996) and Nikiforow et al. (2026)).

subsequent performance.

Employees therefore have strong incentives to maintain a favorable reputation in the eyes of their supervisor (Holmström, 1999; Leary & Kowalski, 1990) – or as Holmström puts it in his Nobel lecture: “If they only knew what the boss wants, they could work on making a good impression in those dimensions.” (Holmström, 2017, p. 1772). Consistent with this notion, prior research shows that employees adjust their behavior when they know that their actions affect how they are evaluated by superiors. For these reasons, employees sometimes refrain from seeking advice (Bauch et al., 2023; Heursen et al., 2023), choose more difficult tasks (Katok & Siemsen, 2011) and prioritize their own work over helping others (Arnold & Tafkov, 2019). Taken together, these findings show how employees change their behavior to create a more favorable impression. As an error provides the supervisor with unfavorable information about the employee, they have strong incentives to adjust their behavior. Although the employee cannot undo the original error, a strong performance signal can help an employee to restore their reputation afterwards.

H1. Employee performance increases after a managerial intervention in response to an error.

2.2 Reputational concerns and performance noise

After being confronted with an error, the most natural way for an employee to restore a damaged reputation would seem to be simply to avoid making another error. Employees may therefore become more cautious, hoping that the supervisor observes this improvement and revises their assessment of the employee’s ability. While avoiding further mistakes is clearly desirable, however, it is not necessarily the most effective way to send a clear signal to the supervisor. One reason is that errors are often rare events, which begs the question how long an employee needs to make no mistakes until this can be considered an improvement? This notion relates to a landmark result from the performance-management literature: A more precise signal provides

more useful information about the underlying ability of an employee (Baker, 1992; Banker & Datar, 1989; Moers, 2006). Consequently, firms select and place greater weight on more informative assessments of performance (Gibbs et al., 2009; Ittner & Larcker, 2002; Krishnan et al., 2005). In turn, performance noise makes it more difficult to draw conclusions from observed actions. For instance, Choi et al. (2020) show that noise complicates learning from experience, while Chan et al. (2023) demonstrate that managers are more likely to promote employees because it is harder to distinguish low effort from unfavorable noise. In summary, some performance dimensions make it more difficult for an employee to make a good impression and to restore their reputation.

2.3 Reputation repair with multiple tasks

In environments where employees are evaluated on a single dimension of performance, an employee seeking to repair a damaged reputation has only one margin along which to improve and should therefore increase effort on that dimension. However, employee performance is measured across multiple dimensions and some of them are more readily observable for the supervisor than others. As effort directed toward a more precise performance dimension provides a stronger signal for a given effort level, employees may have incentives to shift effort towards those dimensions. As the literature on multitasking shows, employees face trade-offs when allocating limited effort across tasks: increasing effort on one dimension can come at the expense of another (Feltham & Xie, 1994; Holmström & Milgrom, 1991; Lu, 2022). A large literature documents such effects in laboratory settings (Hannan et al., 2013; Hannan et al., 2019), while a growing literature in the field shows how changes in incentive structures and performance information induce multitasking (Hong et al., 2018; Kohler et al., 2024; Manthei & Sliwka, 2019). Bringing together reputational concerns and multitasking, Brügger (2011) shows experimentally that career concerns lead employees to shift effort toward more precisely measured tasks because performance on these dimensions provides a stronger signal of their

ability. Following an error, employees may therefore respond not only by increasing overall effort, but also by reallocating effort toward performance dimensions through which they can most effectively restore their reputation.

H2. Employee performance increases more strongly in performance dimensions that provide a more precise signal of employee effort, at the expense of less precise dimensions.

2.4 Reputation repair and learning

The prediction in H2 is based on the assumption that performance improvements primarily require employees to reallocate scarce effort or attention. Errors may, however, also facilitate learning, as suggested by an established literature in behavioral research. The literature on instrumental learning, for instance, argues that errors provide information about cause-and-effect relationships and can thereby improve subsequent decision making and performance (Hull, 1943; Metcalfe, 2017; Simon, 1955). Managers can facilitate this process by encouraging reflection, providing instruction, and creating an environment in which employees openly discuss and learn from mistakes (Bellora-Bienengräber et al., 2023; Fischer et al., 2018; Frese & Keith, 2015; Van Dyck et al., 2005).

If an error conversation increases an employee's knowledge or ability, subsequent performance can improve without requiring the same reallocation of scarce attention across tasks. In particular, when newly acquired knowledge is transferable across activities, performance improvements may extend to other dimensions and thereby dampen negative multitasking spillovers. Eyring (2020), for instance, shows how improvements in one performance dimension can spill over to others when they share a common input. Similarly, employees may learn something from the conversation with their manager that not only helps them avoid future errors but also improves their ability to perform assignments more generally. Thus, even if employees respond strategically to errors as outlined in *H2*, learning may partly offset the

negative spillovers created by effort reallocation. The relative importance of these mechanisms therefore remains an empirical question.

3 Setting and Data

3.1 Firm Setting

The study sample consists of 50 small-to-medium sized firms in the technical service industry. All companies are subcontractors for a large technology firm, hereinafter referred to as *MultiCo*. Contracts between MultiCo and their subcontractors are negotiated annually and contain target agreements on the volume and desired quality of assignment completion (measured through customer feedback). The key metric behind the subcontractors' compensation is the overall volume of completed assignments, however MultiCo can also assign bonuses or penalties depending on whether the firm meets a certain quality target. Firm size ranges between 10 and 190 employees, with the average firm employing 51 workers who are active for MultiCo customers every year. All employees have to pass a mandatory quality exam before working alongside MultiCo's internal technical service division.² It is important to note that even though subcontractors operate in the same technical infrastructure as MultiCo's internal technicians, interaction between employees across or even within firms is very limited, as most assignments are completed by a single person and team members are distributed evenly across districts and regions. MultiCo's field of work is sub-divided into 37 regions, loosely following German postal codes. Technicians are assigned to one specific region and receive their daily schedule directly from the dispatcher of their own firm. Workers are usually tasked with one of two types of assignments, which are *installations* and *troubleshooting*. In the former case, MultiCo has sold a new service or product to the client and the worker now sets it up for the first time

²Unlike MultiCo employees, subcontractors' employees have not completed a state-recognized apprenticeship as electricians and are often new to the daily work of a service technician. In turn, they receive considerably lower wages compared to employees at MultiCo. Part of an employee's total compensation depends on the number of completed assignments. The exact proportion is at the discretion of the individual companies and is not disclosed to MultiCo or third parties.

at the client site. In the latter case, a customer reports an issue with an existing product or service. It is then the task of the technician to investigate and provide a solution at the client site and to ensure that all services are working correctly.³

3.2 Error Management Process

In order to support the technicians before and during their visits, both internal and external technicians have access to a logbook that tracks all prior assignments and employees associated with a client or a specific piece of infrastructure. If an employee makes an error during a work assignment, this can be reported through a dedicated reporting system *only MultiCo employees* have access to. The error management process is depicted in more detail in Figure 1. The errors reported to the manager of the subcontractor usually concern technical issues, such as using wrong materials or impacting the connection of other clients (see Table 9 for more details). While the initial likelihood that an error occurs will depend on employee characteristics and actions, the discovery of the error depends on the work schedules of internal technicians. Therefore, while the timing of the error itself is endogenous and related to employee characteristics, the timing of the confrontation is likely exogenous, as it largely depends on the reporting from MultiCo technicians to the supervisor of the employee who committed the error.⁴ The employees discovering and reporting errors do not personally know the employees who commit the error, even though they work in the same infrastructure. It is therefore unlikely that the reporting of the error is influenced by personal relationships. Given that the quality management of MultiCo is also involved in the reporting process, managers are also less likely to treat employees with the same error in a different way.

Upon submission, the report is reviewed by the subcontractor and MultiCo's quality manage-

³The type and complexity of work assignments tends to be distributed homogeneously across firms, as MultiCo's internal technicians cover the larger share of work assignments with a higher degree of complexity (e.g. corporate clients or advanced technologies).

⁴It is important to note that the error reporting system only concerns errors that are independent from the completion of the work assignment. If the technician fails to complete the work assignment in time, this in itself would not be considered an error that needs to be reported.

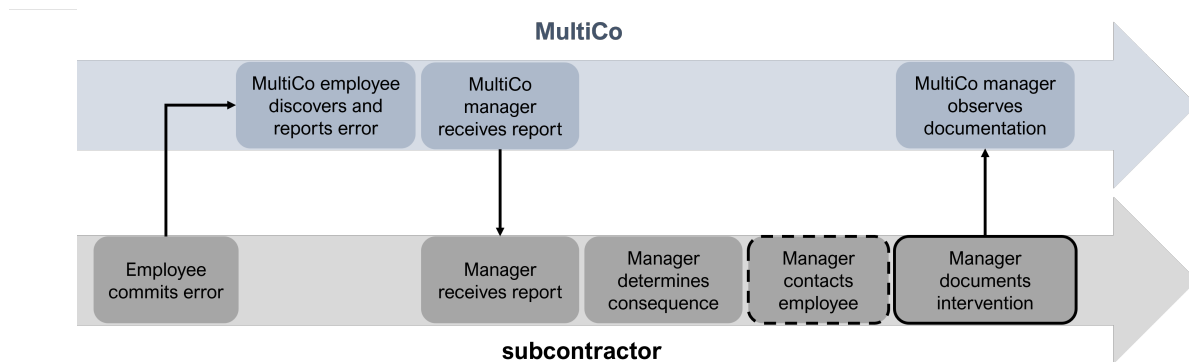


Figure 1: Error Management Process

ment, who jointly decide whether the error report is justified. On average, 80 percent of errors are considered justified by both parties.⁵ Error reports are usually handled directly by the general manager of the subcontractor or a dedicated quality manager. Since the report contains comprehensive documentation and photos, there is limited scope for interpretation at this point. If the error is considered justified by both parties, the manager then contacts the responsible employee and documents the results in the error management system. Managers can choose from different consequences in the documentation, such as on-site training, warnings, and (temporary) suspension. In addition to the category of the consequence, managers need to provide a short written description of their actions. As Figure 1 shows, documenting the details of the intervention is the last step in the error management process. Naturally, employees can only be confronted in the time period between the date of the assessment associated with the error and the date of the documentation by the manager. Based on a sub-sample of 1,029 errors for which the date of the assignment is available, the average processing time is 6.49 weeks ($SD = 12.28$). As Figure 2 shows, most employees can only be confronted close to the time point of the documentation. Specifically, ~40% of all employees are contacted within the last 2.5 weeks prior to documentation, and another 30% are contacted within the last 5 weeks prior to the

⁵While MultiCo has an interest in the adequate handling of errors, they are not allowed to influence a manager's decision or contact an employee directly. However, managers have an incentive to demonstrate to MultiCo that they take these reports seriously and therefore handle them quickly and thoroughly as a sign of proactive compliance.

documentation.

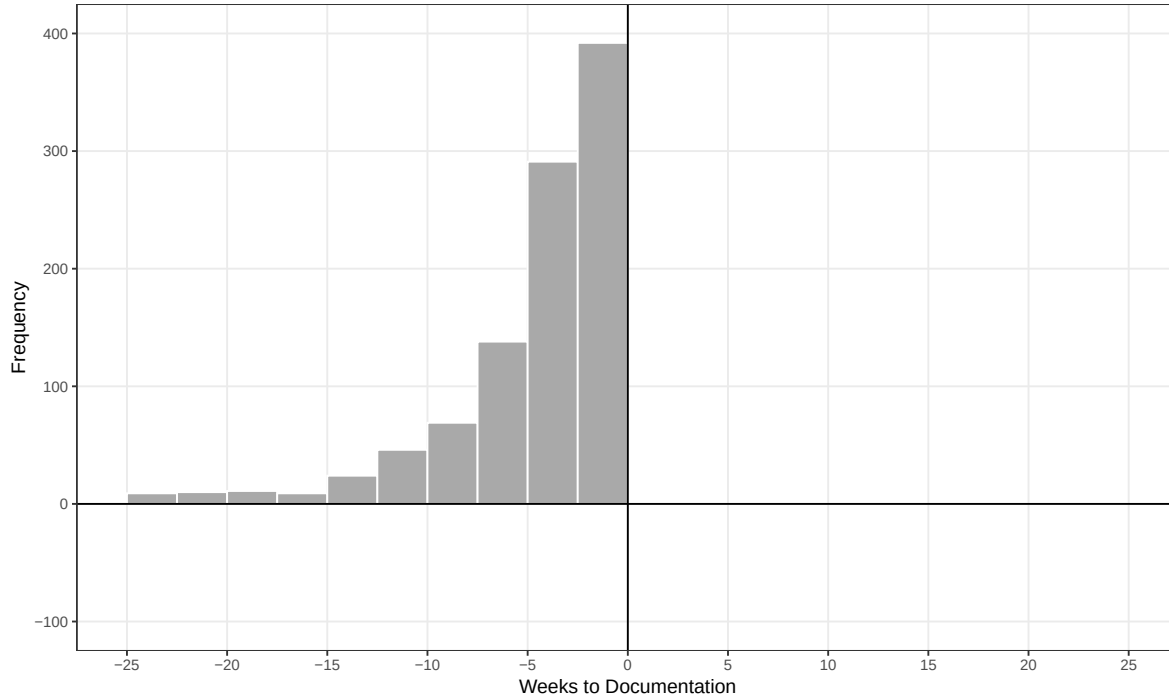


Figure 2: Distribution of Time between Error Report and Manager Documentation

3.3 Data

Table 1 reports summary statistics on key variables for all employees who have been active for at least 3 months according to the available data to provide an accurate representation of employees' performance. Means and standard deviations are calculated on the week-level, except for the employee characteristics. Performance is measured along two dimensions, which are *output quantity* – i.e. the number of completed assignments and the number of days worked – and *output quality* – i.e. the share of successfully completed assignments and customer service ratings. Output quantity is measured directly through the performance records of each employee. On average, employees complete 4.5 assignments per day ($SD = 1.68$) or 19.4 assignments per week ($SD = 7.99$) over a period of 4.12 days of work per week

($SD = 0.66$). Output quality is measured through the customer satisfaction survey, which is sent out via text to each customer. Employees receive an average rating of 4.47 out of 5 stars ($SD = 0.29$). Employees complete most assignments successfully (91%) and on-time (93%) according to the customer evaluation. Given that employees only receive 3.63 evaluations per week, those measures provide a noisier signal of underlying employee ability in comparison to measures of output quantity. Consistent with this, a considerably larger share of the variation in the customer-based measures is residual within-employee variation than for the number of completed assignments (Table 10). In addition to that, satisfaction scores may vary because of customer- or assignment specific factors that are outside of the employee's control. Over the entire observation period, supervisors receive 3.02 justified error reports per employee, which translates to 0.04 reports per week. However, approximately half of all employees do not receive any error report at all.⁶ Therefore, while errors are pervasive across employees, the likelihood of receiving multiple error reports is small. For those reasons, being confronted with an error is a rather unique event, and therefore might have long-lasting effects on future performance in this setting compared to work environments where employees make and discuss mistakes on a more regular basis. Annual turnover across firms ranges between 24% and 32% per year, but the average number of employees per firm remains similar over time, indicating that firms are not facing a net gain or loss in terms of their workforce over time.

3.4 Estimation Procedure

To understand how the confrontation with an error affects an employee's subsequent performance across different dimensions, I use data on performance quantity and quality from 3,265 employees in 50 firms, covering the weekly average number of completed assignments over a period of more than three years, from January 1st, 2021 until April 30th, 2024, which

⁶Figure 4 shows the distribution of errors across employees. While the likelihood of making at least one error that leads to a report is approximately 50%, the figure shows that the number of errors per employee is not equally distributed throughout this group. As the figure shows, 95% of employees are responsible for only 50% of all errors, while the remaining half can be attributed to just 200 employees.

Table 1: Summary Statistics

Variable	Mean	SD
Quantity of Work		
Assignments per Day	4.50	1.68
Days Worked per Week	4.12	0.66
Assignments per Week	19.38	7.99
Quality of Work		
Customer Feedbacks per Week	3.63	1.22
Share of Successfully Completed Assignments	0.91	0.07
Share of Assignments Completed on Time	0.93	0.06
Average Customer Satisfaction Rating	4.47	0.29
Number of Error Reports per Week	0.04	0.09
Employee Characteristics		
Tenure Before Intervention (Weeks)	35.52	35.98
Total Tenure (Weeks)	87.93	56.23

Note: Table reports summary statistics for employees observed for at least 12 weeks (N = 2,405). For quantity and quality measures, employee-level values represent averages across observed weeks. Number of Error Reports denotes the average number of justified error reports per observed employee-week. Tenure Before Intervention measures the number of weeks between an employee's first observation and the manager intervention and is therefore only defined for employees who receive an intervention. Total Tenure measures the observation period between the employee's first and last observed week.

yields 167,256 employee-week observations. Table 8 in the appendix describes the sample selection processes in more detail. The event study is implemented using Equation 1, a standard two-way fixed effects (TWFE) event-study specification. The dependent variable is denoted by $Performance_{it}$, representing for instance the weekly average number of completed work assignments as a measure of output quantity, or the average customer satisfaction rating as a measure of output quality, per employee i in week t . The estimation includes an individual-specific fixed effect, denoted as $Individual_i$ and a time-specific fixed effect, denoted as $Week_t$ which captures any unobserved effects associated with a given week. The leads and lags relative to the week in which the manager has documented the consequence in the error management system are denoted by $Treat_{it}(k)$. Specifically, $k = 0$ corresponds to the week t when the first manager intervention was documented for employee i . The observed time period ranges from $-26 \leq k \leq 26$ with $k = -26$ capturing all time periods at least 26 weeks before the documentation, and $k = 26$ capturing the average values for 26 weeks after the manager intervention and beyond. Using only not-yet-treated observations as control observations reduces the risk that the control group consists of employees who are fundamentally different from those who do receive error reports. The main analysis therefore does not use the full sample of all 3,265 employees, but the sub-sample of 1,165 employees who have received at least one error report. $Treat_{it}(-7)$ is used as the reference point of all coefficients, which is the total time frame that 75% of all errors require to be discovered and processed. The variables of interest are the corresponding coefficients for each time period, denoted by β_k . To separate the more direct effects of the manager intervention from potential post-intervention effects, I compare the coefficients from the pre-treatment period $k \in [-26, -7]$ to those from the *intervention period* which includes all $k \in [-6, -1]$ and those from the *post-intervention period* $k \in [0, 26]$, respectively. Finally, ε_{it} denotes the idiosyncratic error term.

$$Performance_{it} = Individual_i + Week_t + \sum_{\substack{k=-26 \\ k \neq -7}}^{26} \beta_k Treat_{it}(k) + \varepsilon_{it} \quad (1)$$

4 Results

4.1 Output Quantity

Column (1) in Table 2 shows the results for the two periods in relation to the pre-treatment period. In the intervention period, the average number of assignments ($AssignmentsCompleted_{it}$) completed per day increases by 0.254 assignments per day ($p < .001$), which is equal to a 5.35% increase relative to the average number of completed assignments in the pre-treatment period (4.75). For the post-intervention period, the table shows an average performance increase of 0.344 assignments per day ($p < .001$), which corresponds to a 7.24% increase.⁷ Figure 3 shows the graphical results for the effect on the average number of completed assignments following Equation 1. Coefficients and error bars are plotted for each week in relation to the reference period. As the figure shows, no coefficient prior to the reference period is significantly different from zero. After the reference period at $k = -7$, employee performance steadily increases every week – which is in line with a growing share of employees being treated as we move closer to $k = 0$ where all managers have documented their responses to the error (see Figure 2). Column (2) shows the corresponding results for the number of days worked per week ($DaysWorkedWeek_{it}$). During the intervention period, employees work 0.039 additional days per week ($p = .037$), which corresponds to a 0.91% increase relative to the pre-intervention mean of 4.28 days. In the post-intervention period, the number of days worked decreases by 0.033 days per week ($p = .062$), corresponding to a 0.77% decrease relative to the pre-intervention mean. At this point, all employees have been treated and performance has increased by approximately 0.3 to 0.4 assignments per day (6–8%) relative to the reference period and remains at a similar level in subsequent periods. The increase in output quantity appears to be driven primarily by higher output conditional on the same working hours rather than by a sustained increase in

⁷Column (2) of Table 11 shows the combined effect for all time periods after the reference period $k = -7$, while column (3) reports the results for all time periods after the manager's documentation at $k = 0$, which is smaller, but still significant at the 1%-level.

the number of days worked.

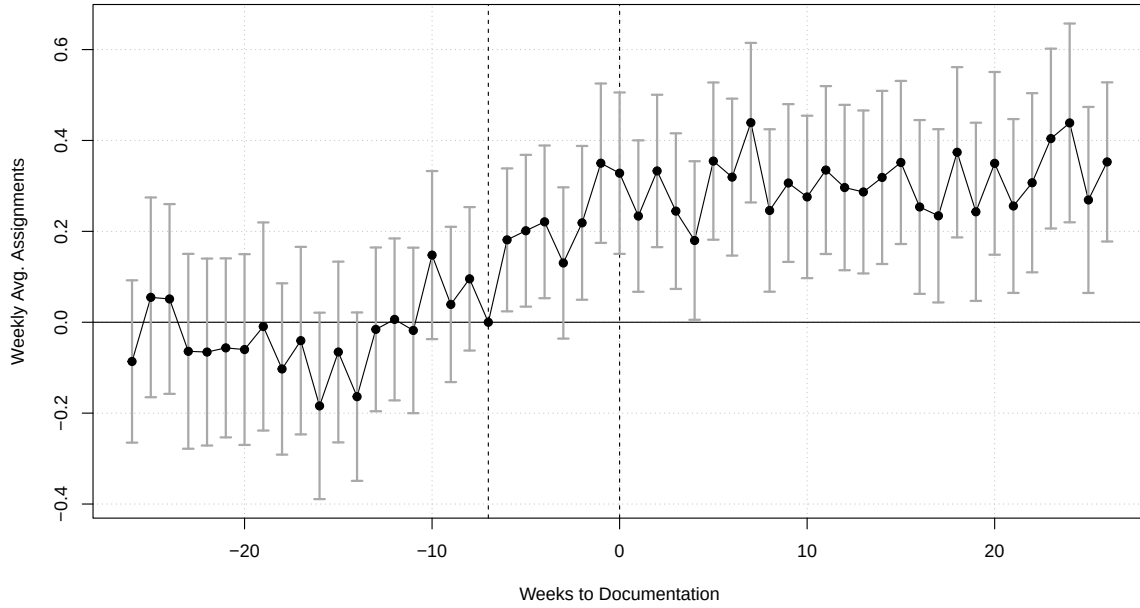


Figure 3: Impact of Manager Intervention on Employee Performance

4.2 Output Quality

While an increase in performance as measured by the number of completed assignments is a positive outcome for the firm, employees might be particularly inclined to improve the quality of their assignments after an error. To study whether the quality of work improves after the confrontation by the manager or - consistent with $H2$ - might even be reduced due to the reallocation of effort, these measures of output quality are studied using data from the customer satisfaction survey. Customer satisfaction ratings are collected in a short survey which is distributed to the customer via text message after the appointment has been completed. The customer feedback survey of the firm consists of three questions, which cover whether the technician was on time for the assignment ($OnTime_{it}$) or not and whether the technician was

able to resolve the issue of the customer ($IssueResolved_{it}$) or not. The third measure concerns the overall impression with the service from the technician. This measure consists of a rating from 1 to 5 stars ($SatisfactionRating_{it}$) and an open text box where customers can provide further comments.⁸ Table 2 shows TWFE-regressions on weekly average punctuality and completion rates, as measured by the customer feedback survey following the same estimation procedure as before. As the table shows, technicians are on time for 93% of the assignments and complete the assignment to the customer's satisfaction in 91% of all cases. The coefficients shown in columns (3) and (4) of Table 2 are close to zero and not significantly different from zero (all $p \geq .155$), which suggests that the quality of work is not affected substantially. Finally, column (5) shows results from the regression on the weekly average customer satisfaction rating measured by rating from 1 to 5 stars.⁹ As the table shows, the average rating is reduced by 0.021 stars ($p = .026$) or 0.45% in the post-intervention period.

To further illuminate the negative effect on customer satisfaction, Table 3 shows a more detailed decomposition of the customer service ratings. Columns (1) to (4) show results for the number of ratings per week per employee ($NumRatings_{it}$), the number of 5 and 1 star ratings ($NumFiveStars_{it}$ and $NumOneStar_{it}$), as well as the average rating for all assignments ($AvgRating_{it}$). On average, employees receive 3.88 ratings per week with an average rating of 4.49 stars, which is reflected by the high share of 5-star ratings (68.47%) and the rather low share of 1-star ratings (3.61%). As Table 3 shows, the overall number of ratings per week as well as the number of 5-star ratings appears to increase slightly, but the potential effect is small and not significantly different from zero (all $p \geq .115$). However, as column 3 shows, the number of 1-star ratings increases significantly both during the intervention period ($Treat \times Intervention$) and afterwards ($Treat \times Post$). In comparison to the pre-intervention mean, the number of 1-star

⁸The wording of the questions are as follows: "The assignment was completed within the announced time frame." and "My issue has been resolved successfully." Both questions are answered with "Yes" or "No". The question regarding overall satisfaction with the technician was answered on a 1-5 stars: "All in all, how satisfied were you with the service from the technician"?

⁹This analysis is focused only on successfully completed assignments, as the customer rating is partially influenced by whether or not the technician has completed the assignment successfully.

Table 2: Impact of Manager Intervention on Employee Performance

	Output Quantity		Output Quality		Customer Service
	Assign-ments Completed	Days Worked	Com-pleted on Time	Issue Resolved	Star Rating
Treat $\times$ Intervention	0.254*** (0.052)	0.039** (0.019)	-0.003 (0.004)	0.001 (0.004)	-0.013 (0.013)
Treat $\times$ Post	0.344*** (0.053)	-0.033* (0.018)	-0.004 (0.003)	-0.001 (0.003)	-0.021** (0.010)
Pre-Intervention Mean	4.75	4.28	0.93	0.91	4.64
Individual FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Observations	95 994	96 378	58 626	59 416	57 833
R ²	0.346	0.271	0.051	0.053	0.077

Note: Table reports estimates from two-way fixed effects regressions including employee and week fixed effects. Columns (1) and (2) capture output quantity. Assignments Completed denotes the average number of MultiCo assignments completed per working day, while Days Worked denotes the number of days worked per week. Columns (3) and (4) capture output quality based on customer feedback. Completed on Time denotes the weekly share of assignments for which the technician arrived at the customer's site within the announced time frame. Issue Resolved denotes the weekly share of assignments reported as successfully completed. Column (5) reports the average customer satisfaction rating for assignments reported as successfully completed. Treat $\times$ Intervention equals one during weeks -6 to -1 relative to the manager's documentation of the intervention. Treat $\times$ Post equals one from week 0 through week 26. The omitted category consists of observations at least seven weeks before documentation. The sample is restricted to employees who receive a justified error report. The number of observations and employees varies across columns because customer-feedback measures are not available for every employee-week. Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

ratings increases by 0.019 ratings per week in the intervention period ($p = .054$; +13.6%), and 0.016 ratings per week in the post-intervention period ($p = .026$; +11.2%). In relative terms, this increase is larger than the increase in the number of completed assignments (5–7%, see Table 2). Overall, this negative effect is reflected in the weekly average rating across all ratings, which is reduced by 0.022 stars ($p = .062$) or 0.49% in the post-intervention period.

Taken together, these findings suggest that the confrontation by the manager led to an improvement in overall employee performance in accordance with hypothesis *H1*. However, in line with hypothesis *H2*, performance improved in the dimension that is more visible to the manager at the cost of a dimension that is more noisily measured. Section 5 investigates the underlying mechanisms of these results in more detail.

Table 3: Impact of Manager Intervention on Customer Satisfaction

	Customer Satisfaction			
	Num. of Ratings	5-Star Ratings	1-Star Ratings	Avg. Rating
Treat × Intervention	0.089 (0.062)	0.042 (0.051)	0.019* (0.010)	−0.013 (0.016)
Treat × Post	0.097 (0.062)	0.021 (0.049)	0.016** (0.007)	−0.022* (0.012)
Pre-Intervention Mean	3.879	2.652	0.143	4.494
Individual FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Observations	59 416	59 416	59 416	58 878
R ²	0.232	0.236	0.059	0.090

Note: The table reports estimates from two-way fixed effects regressions including employee and week fixed effects. Num. of Ratings denotes the total number of customer ratings received by an employee per week. 5-Star Ratings and 1-Star Ratings denote the number of five-star and one-star ratings received per week, respectively. Avg. Rating denotes the average customer satisfaction rating across all ratings on a scale from 1 to 5. Treat × Intervention equals one during weeks -6 to -1 relative to the manager's documentation of the intervention. Treat × Post equals one from week 0 through week 26. The omitted category consists of observations at least seven weeks before documentation. The sample is restricted to employees who receive a justified error report. Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

4.3 Robustness Checks

Section A.1 and Section A.2 in the appendix document robustness checks for the effects on the number of completed assignments and the average customer satisfaction rating. The increase in output quantity is robust to alternative definitions of the intervention and post-intervention periods (Table 11), to the inclusion of additional firm, region, and year-month fixed effects which control for unobserved heterogeneity in time trends and regional characteristics, to including all never-treated employees in the estimation, and to restricting the sample to employees who stay at the firm for at least half a year after the intervention (Table 12).¹⁰ Table 13 further shows similar results for alternative measures of output quantity, such as assignments for different client types and the total number of assignments per week. In addition, a placebo test that assigns randomly drawn intervention dates to never-treated employees shows no comparable increase in output (Figure 7), and all 500 placebo estimates of the post-intervention effect are far below the actual estimate (Figure 8).

Recent research on the canonical two-way fixed-effect (TWFE) estimator has shown that it can yield biased results, especially when treatment adoption is staggered across multiple time periods (see Roth et al., 2023 for a survey). To address these concerns, I re-run the analyses using an imputation estimator (Borusyak et al., 2024; Gardner, 2022; Liu et al., 2024) implemented by Butts and Gardner (2021). For output quantity, the imputation estimator yields a similar effect during the intervention period, but a smaller post-intervention effect (column 5 of Table 12; see Figure 5 for a comparison of the event-study coefficients). The decline in customer satisfaction is robust to alternative period definitions (Table 14), additional fixed effects, and excluding employees who leave the firm, but it becomes smaller and statistically insignificant when never-treated employees are included or when the imputation estimator is used (Table 15). The results for customer satisfaction should therefore be interpreted with more caution than those for output quantity.

¹⁰Section 5.1 examines the likelihood of turnover in response to an error in more detail (see Table 4).

5 Mechanisms

5.1 Effort Reallocation and Learning

While the differential effects across dimensions of employee performance suggest that employees are strategically re-allocating effort from one task to another, there might be additional reasons why treated employees are now able to take on more assignments than before. For instance, it could be that the conversation with the manager has improved their understanding of the job or triggered them to reflect more deeply about their approach to work. If the conversation with the manager led to an increase in the ability of the employee, these employees should be less likely to make future errors compared to similar colleagues who have not been confronted by their manager. If, instead, employees make more mistakes after the confrontation, this would be consistent with a re-allocation of effort away from less toward more readily observable dimensions of performance.

To investigate the likelihood of future errors, the previous estimation strategy is not suitable, as it uses the first error as the treatment. By design, all treated employees have no error record before the treatment, so that no pre-treatment comparison of error rates can be constructed. Instead, I compare treated employees to similar colleagues without a prior error using propensity score matching (PSM) within risk sets. For every employee whose first error is recorded in week t (the index week), the risk set consists of all employees who are active in the same firm and calendar week but have not experienced an error by that date. I restrict the sample to index weeks that permit a complete 26-week follow-up before the end of the observation period. The propensity score $e(X_i) = \Pr(\text{Intervention}_i = 1 \mid X_i)$ is estimated with a logit model, where X_i contains the average number of completed assignments per day, the share of successfully completed assignments, and the average customer satisfaction rating in weeks $t - 26$ to $t - 7$ (the pre-treatment window of the event study), as well as tenure at the index week. Employees with fewer than six observed weeks in this window are excluded. Each treated employee is then matched to the nearest neighbor from the same risk set (1:1 without replacement, caliper

of 0.2 standard deviations of the propensity score). The outcomes are indicators for leaving the firm and for receiving another justified error report within 13 and 26 weeks after the index week. Using the matched sample, I estimate linear probability models of the form

$$Y_{ip} = \alpha_p + \delta \text{Intervention}_i + \varepsilon_{ip} \quad (2)$$

where Y_{ip} denotes the respective outcome of employee i in matched pair p , α_p denotes matched-pair fixed effects, and Intervention_i equals one for the employee whose first error was reported and addressed by the manager. The coefficient δ thus captures the average effect of the intervention on the treated employees (ATT). Because the same employee can serve as a control in several risk sets, standard errors are clustered at the employee level.

Of the 424 first errors that satisfy these restrictions, 343 can be matched within the caliper, while 81 treated employees without a sufficiently similar control are dropped. The sample is therefore considerably smaller than in the main analysis, mainly because of the follow-up requirement and the requirement of sufficient pre-period data. Before matching, treated employees have lower prior quality, lower customer ratings, and shorter tenure than the 11,929 potential control observations, with absolute standardized mean differences of up to 0.22, whereas prior output quantity is already similar. After matching, all absolute standardized mean differences are below 0.1 (maximum: 0.09; see Figure 6 in the appendix).

Table 4 reports the results. Employees who were confronted with their first error are 14.9 percentage points more likely to receive another justified error report within 13 weeks than matched employees without a previous error ($p < .001$; control mean: 18.1%), and 19.5 percentage points more likely within 26 weeks ($p < .001$; control mean: 26.2%). In contrast, the likelihood of leaving the firm does not differ significantly between treated and control employees (2.6 and 3.2 percentage points after 13 and 26 weeks, $p = .132$ and $p = .173$, respectively), suggesting that the intervention does not push employees out of the firm. If the conversation with the manager had improved employees' ability more broadly, treated employees should

make fewer subsequent errors than comparable colleagues. The opposite pattern provides little support for learning as the main explanation for the increase in output and is consistent with a reallocation of effort away from error avoidance. One caveat is that treated employees have revealed an error by construction, whereas matching can only balance observable characteristics. Part of the difference may therefore reflect a persistently higher error propensity of treated employees rather than an effect of the intervention itself.

Since errors are often referred to as an opportunity for learning, it is important to consider which types of errors employees make and whether the nature of the error shapes the structure of the subsequent response. To investigate this heterogeneity, the following analyses differentiate between technical and non-technical errors. Technical errors are the result of insufficient technical knowledge or flawed execution, such as using the wrong technique or materials or accidentally impacting another customer's service. Non-technical errors on the other hand cannot be traced back to a clear deficiency in the technical domain and therefore include cases such as providing the wrong information to the customer or insufficient documentation after a customer visit. Among the 1,165 first errors covered in the analysis, around 58% are technical errors and the remaining 42% can be considered non-technical errors (see Table 9 in the appendix for further details). Given that technical errors can be traced back more easily to a specific part of the assignment process, they should be more useful for discovering blind spots in the employee's skill set and allow the manager to offer more targeted advice. Following the signaling logic on the other hand, technical errors can also be considered more informative of employee ability. That is, technical errors clearly expose a deficiency which can be traced back to the ability of the employee, whereas non-technical errors might be attributed to factors outside of the employee's control. If the latter is the case, output might increase – but at the expense of a reduction in customer service.

Table 4: Matched Estimates of Employee Exit and Subsequent Errors

	Employee Exit		Subsequent Error	
	13 weeks	26 weeks	13 weeks	26 weeks
Manager Intervention	0.026 (0.017)	0.032 (0.024)	0.149*** (0.033)	0.195*** (0.035)
Control Mean	0.044	0.099	0.181	0.262
Observations	686	686	686	686

Note: The table reports linear probability models estimated on a propensity-score-matched sample. Employees experiencing their first error are matched 1:1 (nearest neighbor, without replacement, caliper of 0.2 standard deviations of the propensity score) to employees from the same firm and calendar week who have not previously experienced an error. Matching is based on the average number of completed assignments per day, the share of successfully completed assignments, and the average customer satisfaction rating in weeks $t-26$ to $t-7$ before the index week t , as well as tenure at the index week. Employee exit equals one if the employee's last observed week falls within 13 (26) weeks after the index week. Subsequent error equals one if the employee receives another justified error report within 13 (26) weeks after the index week. Control Mean denotes the outcome mean among matched control employees. All models include matched-pair fixed effects and matching weights. Standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

To investigate whether the response differs by error type, Table 5 reports TWFE regressions separately for employees who committed technical and non-technical errors. Columns (1) and (2) examine the number of completed assignments, while columns (4) and (5) examine the number of one-star customer ratings. To test these differences directly, columns (3) and (6) report triple-difference (TD) specifications for the two respective outcomes, implemented as:

$$\begin{aligned}
\text{Outcome}_{it} = & \text{Individual}_i + \text{Week}_{t \times g} + \beta_1 \text{Intervention}_{it} + \beta_2 \text{Post}_{it} \\
& + \gamma_1 (\text{Intervention}_{it} \times \text{Technical}_i) + \gamma_2 (\text{Post}_{it} \times \text{Technical}_i) + \varepsilon_{it}
\end{aligned} \tag{3}$$

where Outcome_{it} denotes either the average number of completed assignments per working day or the weekly number of one-star ratings. Technical_i equals one for employees whose first error is classified as technical and zero for employees with a non-technical error. The TD

specifications additionally allow week fixed effects to differ between the two error types g , as indicated by $Week_{t \times g}$. The parameters of interest are γ_1 and γ_2 . The former captures whether the change during the intervention period $k \in [-6, -1]$ differs following technical relative to non-technical errors, while the latter captures the corresponding differential response during the post-intervention period including all periods $k \in [0, 26]$. Thus, positive values indicate a larger increase in the respective outcome following technical errors.

As Table 5 shows, the increase in output quantity is very similar after technical and non-technical errors, both during the intervention period (0.249 vs. 0.269 assignments per day) and afterwards (0.356 vs. 0.326), and the triple-difference estimates are small and statistically insignificant ($p = .858$ and $p = .794$). In contrast, the increase in 1-star ratings is concentrated among employees whose first error was technical: Their number of 1-star ratings increases by 0.032 ratings per week during the intervention period ($p = .012$) and by 0.018 afterwards ($p = .043$), whereas the corresponding estimates for non-technical errors are small and insignificant (both $p \geq .269$). The differential effect is marginally significant during the intervention period (0.036, $p = .072$), but not in the post-intervention period ($p = .709$). Taken together, the type of error does not affect how strongly employees increase their output, but the deterioration in customer service is more pronounced after errors that are more informative about an employee's technical ability.

5.2 Reputational Concerns

Following the prior literature on reputational concerns, some employees may have stronger incentives to increase performance after being confronted with an error. First, employees with lower prior performance are likely to have a less favorable standing with their supervisor and therefore generally have a higher risk of being terminated. An error attributable to the employee provides additional negative information and may therefore push the supervisor's assessment closer to, or below, the threshold where the supervisor believes that the employee is not capable

Table 5: Impact of Manager Intervention by Error Type

	Assignment Quantity			1-Star Ratings		
	Techni- cal	Non- Technical	Triple- Diff.	Techni- cal	Non- Technical	Triple- Diff.
Treat × Interv.	0.249*** (0.065)	0.269*** (0.086)	0.269*** (0.086)	0.032** (0.013)	−0.004 (0.016)	−0.004 (0.016)
Treat × Post	0.356*** (0.064)	0.326*** (0.094)	0.326*** (0.094)	0.018** (0.009)	0.013 (0.011)	0.013 (0.011)
Treat × Interv. × Technical			−0.019 (0.108)			0.036* (0.020)
Treat × Post × Technical			0.030 (0.114)			0.005 (0.014)
Pre-Intervention Mean	4.71	4.82	4.75	0.14	0.15	0.14
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Type × Week FE	No	No	Yes	No	No	Yes
Observations	63 730	32 264	95 994	39 094	20 322	59 416
R ²	0.358	0.324	0.348	0.064	0.055	0.061

Note: The table reports estimates from two-way fixed effects regressions examining heterogeneous responses to manager interventions by error type. Columns (1)-(3) use the average number of assignments completed per working day as the dependent variable, while columns (4)-(6) use the weekly number of one-star customer ratings. Technical errors include errors involving the technique or materials used, documentation, incorrect connections, or effects on unrelated customers. Non-technical errors include customer service complaints, incorrect information provided to customers, missing sources of error, and assignments not processed correctly. Columns (3) and (6) estimate the differential response following technical relative to non-technical errors. Treat × Intervention equals one during weeks -6 to -1 relative to the manager's documentation of the intervention. Treat × Post equals one from week 0 through week 26. The omitted category consists of observations at least seven weeks before documentation. The triple-difference specifications include error-type-specific week fixed effects. All regressions include employee fixed effects. Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

enough to remain with the firm. In contrast, high-performing employees have accumulated a more favorable reputation over time and thus have a larger “buffer” against negative shocks. Therefore, one negative signal should have fewer implications for their standing in the firm, particularly in a setting in which employees primarily face downside career risk rather than substantial opportunities for upward mobility.

Second, employees with shorter tenure are still in the process of establishing a reputation for themselves (Hermalin & Weisbach, 2017; Holmström, 1999). When supervisors have observed only a limited number of prior performance signals, their beliefs about an employee’s underlying ability remain relatively uncertain. A new performance signal, whether positive or negative, therefore receives greater weight in updating the belief of the manager. For employees with longer tenure, supervisors have accumulated substantially more information about past performance, making their assessment of employee ability more precise. A single negative signal in the form of an error should consequently have a smaller effect on their assessment. Short-tenure employees should therefore face stronger incentives to repair their reputation following an error.

To investigate these predictions, Table 6 shows TWFE regressions on the number of completed assignments for employees with low and high prior performance (columns 1 and 2), and low and high tenure (columns 4 and 5), respectively. To study these differential effects more directly, columns (3) and (6) show results from a TD regression, following the same structure as Equation 3, where the interaction terms γ_1 and γ_2 now include a dummy variable (Low_i) which is equal to 1 for all employees with below-median prior assignment performance for the analyses on prior performance (and 0 otherwise), and equal to 1 for all employees with less than one year of tenure for the analysis on prior tenure (and 0 otherwise).

As Table 6 shows, employees with lower prior performance respond more strongly to the intervention compared to those with high prior performance. During the intervention period, high-performing employees increase their output by 0.169 assignments per day ($p = .028$),

corresponding to an increase of approximately 3.0% relative to their pre-intervention mean of 5.63 assignments. In contrast, low-performing employees increase output by 0.392 assignments per day ($p < .001$), or approximately 11.2% relative to their pre-intervention mean of 3.51 assignments. This difference is also statistically significant in the triple-difference specification (0.223, $p = .030$). The divergence becomes even stronger in the post-intervention period. High-performing employees show essentially no persistent change in output, with an estimated effect of -0.009 assignments per day ($p = .900$), whereas low-performing employees increase output by 0.739 assignments per day ($p < .001$). The corresponding triple-difference estimate is statistically significant (0.748, $p < .001$).

A similar pattern emerges for prior tenure: Employees with at least one year of tenure prior to the intervention increase output by 0.092 assignments per day during the intervention period ($p = .233$), corresponding to approximately 1.9% of their pre-intervention mean of 4.84 assignments. Employees with shorter tenure increase output by 0.341 assignments per day ($p < .001$), or approximately 7.4% relative to their pre-intervention mean of 4.58 assignments. The difference between the two groups is statistically significant in the triple-difference specification (0.249, $p = .013$). In the post-intervention period, long-tenure employees again show no positive performance response, with an estimated effect of -0.091 assignments per day ($p = .283$), while short-tenure employees increase output by 0.547 assignments per day ($p < .001$), corresponding to approximately 11.9% of their pre-intervention mean. This differential post-intervention response is statistically significant (0.637, $p < .001$). Overall, the results are consistent with stronger performance responses among employees with a fragile or less established reputation.¹¹

¹¹A potential explanation for these results could be that managers handle confrontations with less-experienced or low-performing employees differently than confrontations with their more established colleagues. Table 16 and Table 17 in the appendix compare the managers' type of response and the effort that goes into the documentation. The type of response does not differ significantly between groups at the 5% level. Managers write somewhat shorter statements for low-performing employees and use the pre-filled standard statement less often for low-tenure employees, but overall, the comparisons do not indicate that managers systematically invest more effort in the groups that show the strongest response.

Table 6: Impact of Manager Intervention by Prior Performance and Tenure

	Prior Performance			Prior Tenure		
	High Perf.	Low Perf.	Triple-Diff.	High Tenure	Low Tenure	Triple-Diff.
Treat × Intervention	0.169** (0.076)	0.392*** (0.069)	0.169** (0.076)	0.092 (0.077)	0.341*** (0.064)	0.092 (0.077)
Treat × Post	−0.009 (0.070)	0.739*** (0.081)	−0.009 (0.070)	−0.091 (0.084)	0.547*** (0.072)	−0.091 (0.084)
Treat × Interv. × LowPerf			0.223** (0.103)			
Treat × Post × LowPerf			0.748*** (0.107)			
Treat × Interv. × LowTenure						0.249** (0.100)
Treat × Post × LowTenure						0.637*** (0.111)
Pre-Intervention Mean	5.63	3.51	4.75	4.84	4.58	4.75
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Group × Week FE	No	No	Yes	No	No	Yes
Observations	45 359	33 898	79 257	30 688	65 306	95 994
R ²	0.276	0.296	0.352	0.353	0.347	0.349

Note: The table reports estimates from two-way fixed effects regressions on the average number of assignments completed per working day. Columns (1) and (2) split employees by their average assignment performance prior to the error, while Column (3) estimates the differential response of low- relative to high-performing employees. Columns (4) and (5) split employees by prior tenure. High Tenure includes employees observed for at least 52 weeks prior to documentation, while Low Tenure includes employees observed for less than 52 weeks. Column (6) estimates the differential response of low- relative to high-tenure employees. Treat × Intervention equals one during weeks -6 to -1 relative to the manager's documentation of the intervention. Treat × Post equals one from week 0 onward. The omitted category consists of observations at least seven weeks before documentation. The triple-difference specifications include group-specific week fixed effects. All regressions include employee fixed effects. Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

In addition to employee characteristics such as performance and tenure, the likelihood of termination can also be influenced by the external labor market. As a literature in labor economics shows, in regions with higher unemployment it becomes more attractive to replace workers rather than providing training to them (Arellano-Bover, 2022; Brunello, 2009; Majumdar, 2007). Therefore, employees in regions with higher unemployment may be at a greater threat of being replaced due to bad performance. To explore heterogeneity in the role of this risk, the following analyses combine German census data on local unemployment (DeStatis, 2025) with the regions provided by MultiCo. While employees can take on assignments in different regions, they are usually assigned to a single region which is why they are allocated to the regions they completed the most assignments in. Each region is then categorized as “high” or “low” unemployment using a median split. The dummy *HighUnemployment_i* is equal to 1 for all employees who are primarily assigned to a region with above-median unemployment, and 0 otherwise. Using the same estimation procedure as before, Table 7 shows that while the effects are numerically larger for employees working in areas with higher unemployment rates (columns 1 and 2), the triple-difference estimation does not show significant differences between both areas (column 3; $p = .529$ for the intervention period and $p = .419$ for the post-intervention period).

Table 7: Impact of Manager Intervention by Regional Unemployment

	Regional Unemployment		
	Low Unemployment	High Unemployment	Triple-Diff.
Treat × Interv.	0.206*** (0.074)	0.271*** (0.072)	0.206*** (0.074)
Treat × Post	0.296*** (0.081)	0.382*** (0.070)	0.296*** (0.081)
Treat × Interv. × HighUnemp.			0.065 (0.103)
Treat × Post × HighUnemp.			0.086 (0.107)
Pre-Intervention Mean	5.27	4.34	4.73
Individual FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Group × Week FE	No	No	Yes
Observations	44 512	50 319	94 831
R ²	0.358	0.327	0.352

Note: The table reports estimates from two-way fixed effects regressions on the average number of assignments completed per working day. Columns (1) and (2) split employees according to regional unemployment. Low Unemployment includes employees working in regions with below-median unemployment, while High Unemployment includes employees working in regions with above-median unemployment. Column (3) estimates the differential response of employees in high- relative to low-unemployment regions. Treat × Intervention equals one during weeks -6 to -1 relative to the manager's documentation of the intervention. Treat × Post equals one from week 0 through week 26. The omitted category consists of observations at least seven weeks before documentation. The triple-difference specification includes unemployment-group-specific week fixed effects. All regressions include employee fixed effects. Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

6 Cost-Benefit Analysis

Finally, it is important to consider whether the positive effects on the number of assignments outweigh the costs of the reduction in customer service quality. To calculate the magnitude of these countervailing effects, I use proprietary data on assignment revenue and profits per MultiCo customer, and combine these data with industry benchmarks on cost structures in the technical service industry. To begin with, the profit increase for the subcontractor is derived from the post-intervention increase in weekly assignments shown in column (4) of Table 13 multiplied by the estimated profit per assignment, which is derived from the revenue per assignment and the firms' cost structure.¹² The compensation per assignment varies across firms and cannot be disclosed publicly. Instead, all monetary amounts are expressed in relation to the estimated profit from one standard assignment. While the additional profits associated with the increase in assignments completed are realized directly by the subcontractor, the decrease in customer service quality is more likely to affect MultiCo, as customers may be reluctant to extend their contract or cancel it entirely if possible. The estimated profit loss can be calculated from the post-intervention increase in 1-star reviews in Table 3, the probability of a customer cancelling their contract in case of a 1-star rating and the total profit per customer for MultiCo over the entire customer life-cycle (which was obtained from internal data from MultiCo). Even though MultiCo cannot pass on these costs directly to the subcontractor, they might account for such risk in future negotiations, for example by reducing the amount of assignments or offering less revenue per assignment. Assuming that MultiCo passes on these costs to the subcontractor, the overall net profit from the intervention can therefore be estimated as follows:

¹²Since the majority of companies in the sample are not required by law to publish their income statements, cost structures cannot be estimated individually but must be approximated based on local industry standards. I therefore approximate the average cost ratio per unit of output using administrative data on the local construction and installation sector. The margin factor, which determines the profit per assignment, is therefore defined as the net operating surplus minus personnel costs, divided by the total revenue per assignment.

$$\Delta \text{Profit} = \underbrace{\text{Effect}_{\text{assignment}} \times \text{Profit}_{\text{assignment}}}_{\text{Profit Gain from Effect on Assignments}} - \underbrace{\text{Effect}_{\text{rating}} \times P(\text{cancel} \mid 1\text{-star}) \times \text{Profit}_{\text{client}}}_{\text{Profit Loss from Effect on 1-Star Ratings}} \quad (4)$$

To illustrate the calculation behind Equation 4, I set the profit per assignment to 10 USD. With 1.373 additional assignments per week in the post-intervention period (column 4 of Table 13), this yields an additional profit of 13.73 USD per week per employee. While a 1-star rating reflects a strong dissatisfaction with the service, the switching costs in the industry MultiCo operates in are relatively high. Assuming that a 1-star rating doubles the likelihood that the customer cancels the contract, each additional 1-star rating implies an expected profit loss of around 402 USD. Given that the number of 1-star ratings increases by 0.016 ratings per week in the post-intervention period (Table 3), the associated profit loss for MultiCo equals 6.43 USD per week per employee. In summary, even though the reduction in customer service quality is economically meaningful, the profit associated with the additional assignments is around 2.1 times as large.

7 Conclusion

This paper examines how employees respond when their manager confronts them with a workplace error. Using performance and error records of more than 3,200 service technicians in 50 subcontracting firms, I find that output quantity increases after the manager's intervention: Employees complete about 5% more assignments per day during the intervention period and about 7% more in the following six months, mainly because they complete more assignments per working day rather than working more days. Output quality, in contrast, does not improve. Punctuality and completion rates remain unchanged, while customer satisfaction declines slightly, reflected in an increase in 1-star ratings of 11–14%. Employees who experience an intervention are also more likely to receive another error report within three and six months

than matched colleagues without a prior error, while their likelihood of leaving the firm does not differ significantly. Together, these patterns provide little support for broad learning and are more consistent with a reallocation of effort toward the performance dimension that is most visible to the supervisor. The increase in output is concentrated among employees with lower prior performance and shorter tenure, whose standing with the supervisor is likely less secure, which is consistent with reputation repair as the underlying motive. The output response does not differ by regional unemployment or by the type of error, although the increase in 1-star ratings is concentrated among employees whose first error was technical. An illustrative cost-benefit calculation suggests that the value of the additional output is about twice as large as the expected cost of lower customer satisfaction.

These findings have implications for the design of accountability systems. Holding employees accountable for their errors can increase effort, but employees appear to direct this effort toward the most precisely measured dimension of performance rather than toward the dimension in which the error occurred. Firms that want accountability to improve the quality of work may therefore need to make quality improvements more visible, for instance by providing more frequent feedback on quality and customer service, or to frame error conversations around learning rather than evaluation.

This paper has several limitations. First, while the timing of the intervention largely depends on when MultiCo technicians discover an error, it is not randomly assigned, and the event-study approach cannot rule out all alternative explanations for the observed effects. Second, the matching analysis can only balance observable characteristics. Since treated employees have revealed an error by construction, their higher likelihood of subsequent errors may partly reflect a persistently higher error propensity. Third, the decline in customer satisfaction is small and less robust than the increase in output, as it becomes insignificant when never-treated employees are included or an imputation estimator is used. Fourth, I do not observe the content of the conversations, the time employees spend per assignment, or how employees perceive

the intervention, which limits how precisely the underlying mechanism can be identified. Finally, the cost-benefit calculation relies on assumptions about profit margins and cancellation probabilities and should be read as illustrative. Future research could study error management in settings where errors are more frequent or more visible, compare different managerial responses to errors across firms, and examine how employees perceive being held accountable for their mistakes.

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Appendix

Tables

Table 8: Sample Selection Process

Description	Observations	Employees
Performance records (initial data set)	212,356	3,306
Exclude back-office workers and sub-sub contractors	212,128	3,265
Exclude employee-weeks without any completed assignments	167,256	3,265
Error reports (initial data set)	10,084	2,133
Exclude back-office workers and sub-sub contractors	9,882	2,111
Exclude cases of misconduct	9,269	1,997
Exclude consequences without error report	4,476	1,215
Only keep the first error per employee	1,161	1,161

Table 9: Relative Share of Error Types

Error Description	Total	Relative Share
Technical Errors	674	58.06%
Wrong technique or materials used		21.41%
Incorrect connection established		18.83%
Implausible measurements		11.84%
Impacted connection of other customers		5.96%
Non-Technical Errors	487	41.94%
Unspecified source of error		23.55%
Insufficient or wrong assignment documentation		7.91%
Incorrect information provided to customer		5.70%
Insufficient customer service		3.15%
Assignment not processed correctly		1.63%
Total	1,161	100.00%

Table 10: Residual Variation in Performance Measures

Performance Measure	Residual Variance	Observations
Assignments Completed	0.655	95,994
Customer Issue Resolved	0.947	59,416
Assignment Completed on Time	0.949	58,626
Customer Satisfaction Rating	0.910	58,878

Note: The table reports residual variance in weekly performance measures. Each outcome is standardized to a mean of zero and a standard deviation of one before estimation. Residuals are obtained from separate regressions including employee and week fixed effects. Higher residual variance indicates that a larger share of variation occurs within employees over time after removing persistent employee differences and common week-level shocks. Residual variation captures both transitory changes in performance and measurement noise and should therefore not be interpreted as pure measurement error.

A.1 Robustness Checks: Output Quantity

Table 11: Alternative Definitions of the Post-Intervention Period (Output Quantity)

	Separate Analysis	Including Intervention	Post Only
Treat × Intervention	0.254*** (0.052)		
Treat × Post	0.344*** (0.053)	0.321*** (0.048)	0.263*** (0.045)
Observations	95 994	95 994	95 994
R ²	0.346	0.346	0.346

Note: The table reports estimates from two-way fixed effects regressions of the average number of assignments completed per working day on alternative definitions of the intervention and post-intervention periods. All regressions include employee and week fixed effects. Column (1) estimates the intervention period and post-intervention period separately: Treat × Intervention equals one during weeks -6 to -1 relative to the manager's documentation, while Treat × Post equals one from week 0 through week 26. In Column (2), the treatment indicator equals one for all periods after week -7, thereby combining the intervention and post-intervention periods. In Column (3), the treatment indicator equals one from week 0 onward, while all periods before documentation constitute the reference period. Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 12: Impact of Manager Intervention on Output Quantity (Robustness)

	Assignments Completed				
	(1)	(2)	(3)	(4)	(5)
Treat × Intervention	0.254*** (0.052)	0.254*** (0.052)	0.196*** (0.051)	0.284*** (0.062)	0.306*** (0.054)
Treat × Post	0.344*** (0.053)	0.344*** (0.053)	0.202*** (0.052)	0.384*** (0.060)	0.148* (0.077)
Individual FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Firm FE	No	Yes	Yes	Yes	No
Region FE	No	Yes	Yes	Yes	No
Year-Month FE	No	Yes	Yes	Yes	No
Including Never-Treated	No	No	Yes	No	Yes
Excluding Exits	No	No	No	Yes	No
Gardner Two-Stage DiD	No	No	No	No	Yes
Observations	95 994	95 994	166 943	84 745	147 851
R ²	0.346	0.346	0.366	0.326	0.002

Note: Table shows estimates from two-way fixed effect regressions on the weekly average number of MultiCo assignments per employee per day. Treat × Intervention is a binary indicator equal to 1 for all periods k between and including -6 and -1, and 0 otherwise. Treat × Post is a binary indicator equal to 1 for all periods k greater than or equal to 0, and 0 otherwise. Column (1) reports the baseline specification including employee and week fixed effects, where week is a running indicator for the current week in the observation period. Column (2) additionally includes firm, region, and year-month fixed effects. Column (3) uses the specification in Column (2) and additionally includes never-treated employees. Column (4) restricts the sample to employees who are observed for at least 26 weeks after the manager's documentation. Column (5) shows estimates from an imputation estimator implemented by Butts & Gardner (2021). Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 13: Impact of Manager Intervention on Alternative Measures of Output Quantity

	Client Type			Work Volume	
	MultiCo	Non-MultiCo	All Clients	Weekly Assignments	Days Worked
Treat × Intervention	0.254*** (0.052)	0.259*** (0.061)	0.476*** (0.083)	1.305*** (0.255)	0.039** (0.019)
Treat × Post	0.344*** (0.053)	0.124* (0.067)	0.442*** (0.089)	1.373*** (0.263)	−0.033* (0.018)
Pre-Intervention Mean	4.75	4.74	9.29	20.93	4.28
Individual FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Observations	95 994	95 994	95 994	95 994	96 378
R ²	0.346	0.462	0.456	0.334	0.271

Note: The table reports estimates from two-way fixed effects regressions using alternative measures of employee output. Columns (1) to (3) distinguish assignments by client type. MultiCo denotes the average number of MultiCo assignments completed per working day, Non-MultiCo denotes the corresponding average for other clients, and All Clients includes assignments for all clients. Column (4) reports the total number of assignments completed per week. Column (5) reports the number of days per week on which the employee completed at least one assignment. Treat × Intervention equals one during weeks -6 to -1 relative to the manager's documentation, while Treat × Post equals one from week 0 through week 26. The omitted category consists of observations at least seven weeks before documentation. All regressions include employee and week fixed effects. Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

A.2 Robustness Checks: Customer Service Ratings

Table 14: Alternative Definitions of the Post-Intervention Period for Customer Ratings

	Separate Analysis	Including Intervention	Post Only
Treat × Intervention	−0.013 (0.013)		
Treat × Post	−0.021** (0.010)	−0.019** (0.009)	−0.017** (0.009)
Observations	57 833	57 833	57 833
R ²	0.077	0.077	0.077

Note: The table reports estimates from two-way fixed effects regressions of the weekly average customer satisfaction rating for successfully completed assignments on alternative definitions of the intervention and post-intervention periods. Customer ratings range from 1 to 5 stars, with higher values indicating greater customer satisfaction. All regressions include employee and week fixed effects. Column (1) estimates the intervention period and post-intervention period separately: Treat × Intervention equals one during weeks -6 to -1 relative to the manager's documentation, while Treat × Post equals one from week 0 through week 26. In Column (2), the treatment indicator equals one for all periods after week -7, thereby combining the intervention and post-intervention periods. In Column (3), the treatment indicator equals one from week 0 onward, while all periods before documentation constitute the reference period. Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 15: Robustness Checks for Customer Satisfaction

	Average Rating				
	(1)	(2)	(3)	(4)	(5)
Treat × Intervention	−0.013 (0.013)	−0.013 (0.013)	−0.010 (0.013)	−0.023 (0.016)	−0.006 (0.015)
Treat × Post	−0.021** (0.010)	−0.021** (0.010)	−0.015 (0.009)	−0.019* (0.011)	0.002 (0.013)
Individual FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Firm FE	No	Yes	Yes	Yes	No
Region FE	No	Yes	Yes	Yes	No
Year-Month FE	No	Yes	Yes	Yes	No
Including Never-Treated	No	No	Yes	No	Yes
Excluding Exits	No	No	No	Yes	No
Gardner Two-Stage DiD	No	No	No	No	Yes
Observations	57 833	57 833	90 814	51 106	61 699
R ²	0.077	0.077	0.081	0.074	0.000

Note: Table shows estimates from two-way fixed effect regressions on the average customer satisfaction rating for all successfully completed assignments in a given week. Treat × Intervention is a binary indicator equal to 1 for all periods k between and including -6 and -1, and 0 otherwise. Treat × Post is a binary indicator equal to 1 for all periods k greater than or equal to 0, and 0 otherwise. Column (1) reports the baseline specification including employee and week fixed effects, where week is a running indicator for the current week in the observation period. Column (2) additionally includes firm, region, and year-month fixed effects. Column (3) uses the specification in Column (2) and additionally includes never-treated employees. Column (4) restricts the sample to employees who are observed for at least 26 weeks after the manager's documentation. Column (5) shows estimates from an imputation estimator implemented by Butts & Gardner (2021). Robust standard errors clustered at the employee level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

A.3 Manager Responses by Employee Group

Table 16: Manager Responses to High- and Low-Performing Employees

	High (N=438)		Low (N=425)		Diff. in Means	p
	Mean	Std. Dev.	Mean	Std. Dev.		
Manager Intervention						
Share Intervention	0.83	0.38	0.82	0.39	−0.01	0.756
Share Remote Instruction	0.65	0.48	0.63	0.48	−0.02	0.492
Share Onsite Instruction	0.07	0.25	0.06	0.24	−0.01	0.762
Share Warning	0.11	0.32	0.13	0.34	0.02	0.373
Manager Effort						
Statement Length	155.42	196.32	126.60	164.58	−28.82	0.034
Share Standard Statement	0.47	0.50	0.51	0.50	0.04	0.283

Note: The table compares managerial responses to employees with above-median (High) and below-median (Low) average assignment performance prior to their first error. Share Intervention denotes the share of error reports for which the manager documented a response. Share Remote Instruction, Share Onsite Instruction, and Share Warning denote the share of error reports associated with the respective managerial response. Statement Length denotes the number of characters in the manager's written statement documenting the intervention. Share Standard Statement denotes the share of cases in which the manager used the pre-filled standard statement in the error management system rather than providing an individual written description. The final column reports p-values from tests of differences between high- and low-performing employees.

Table 17: Manager Responses by Prior Tenure

	High (N=251)		Low (N=824)		Diff. in Means	p
	Mean	Std. Dev.	Mean	Std. Dev.		
Manager Intervention						
Share Intervention	0.82	0.38	0.82	0.39	0.00	0.949
Share Remote Instruction	0.68	0.47	0.62	0.49	−0.06	0.075
Share Onsite Instruction	0.05	0.21	0.08	0.27	0.03	0.069
Share Warning	0.10	0.29	0.12	0.33	0.03	0.180
Manager Effort						
Statement Length	122.79	178.30	147.53	180.22	24.75	0.083
Share Standard Statement	0.55	0.50	0.45	0.50	−0.10	0.009

Note: The table compares managerial responses to employees with high and low prior tenure. High-tenure employees are observed for at least 52 weeks before their first error, while low-tenure employees are observed for less than 52 weeks before their first error. Share Intervention denotes the share of error reports for which the manager documented a response. Share Remote Instruction, Share Onsite Instruction, and Share Warning denote the share of error reports associated with the respective managerial response. Statement Length denotes the number of characters in the manager's written statement documenting the intervention. Share Standard Statement denotes the share of cases in which the manager used the pre-filled standard statement in the error management system rather than providing an individual written description. The final column reports p-values from tests of differences between high- and low-tenure employees.

Figures

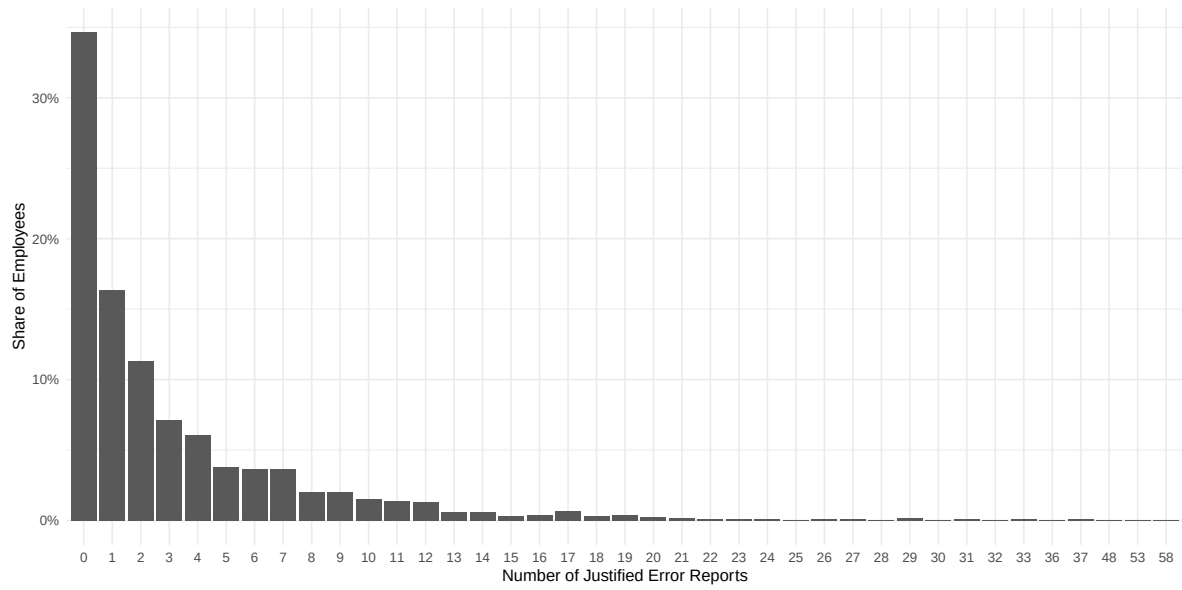


Figure 4: Distribution of Errors Across Employees

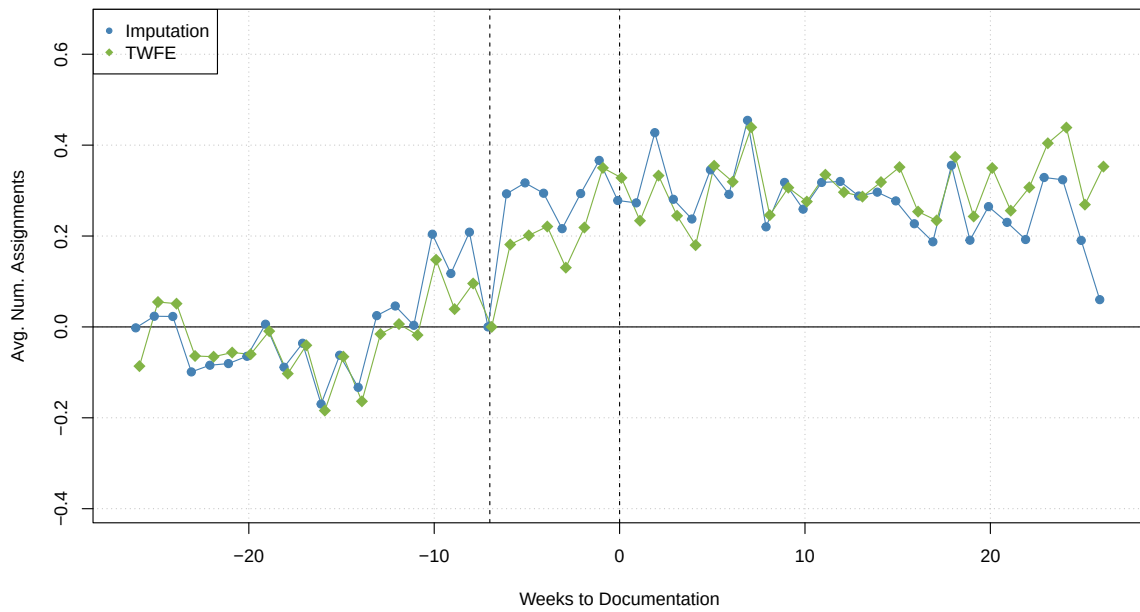


Figure 5: Impact of Manager Intervention on Assignment Quantity: TWFE vs. Imputation Estimator

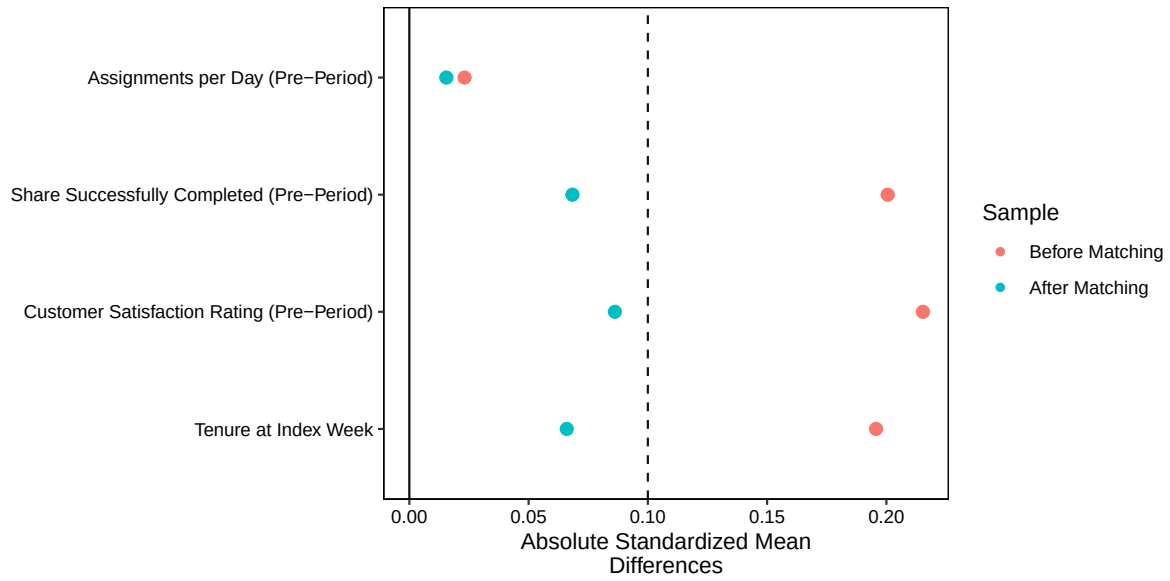


Figure 6: Covariate Balance Before and After Propensity Score Matching

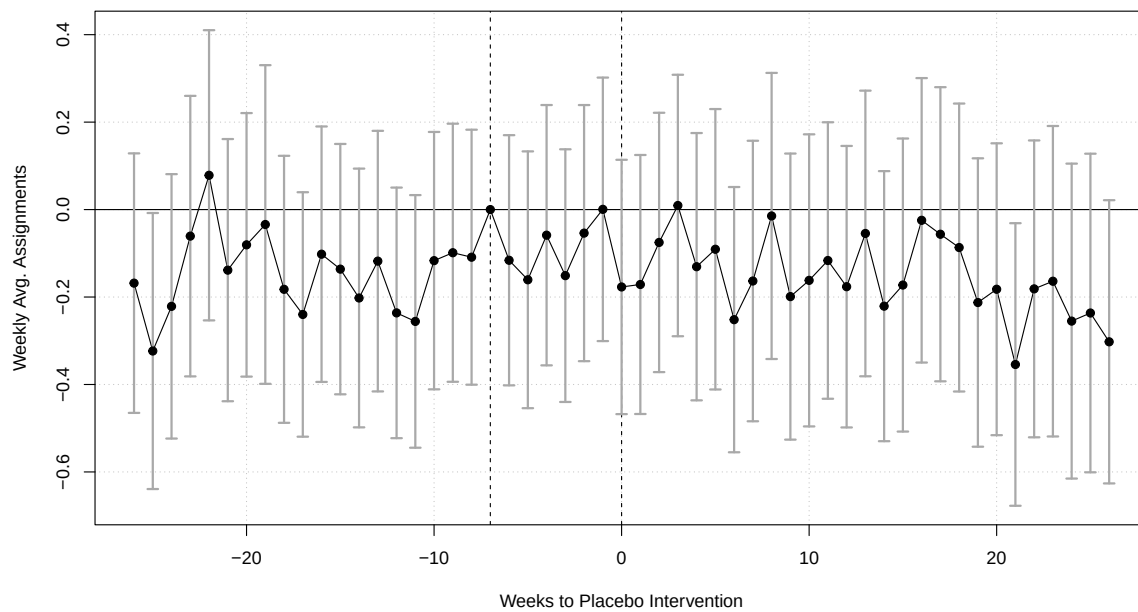


Figure 7: Placebo Event Study for Never-Treated Employees

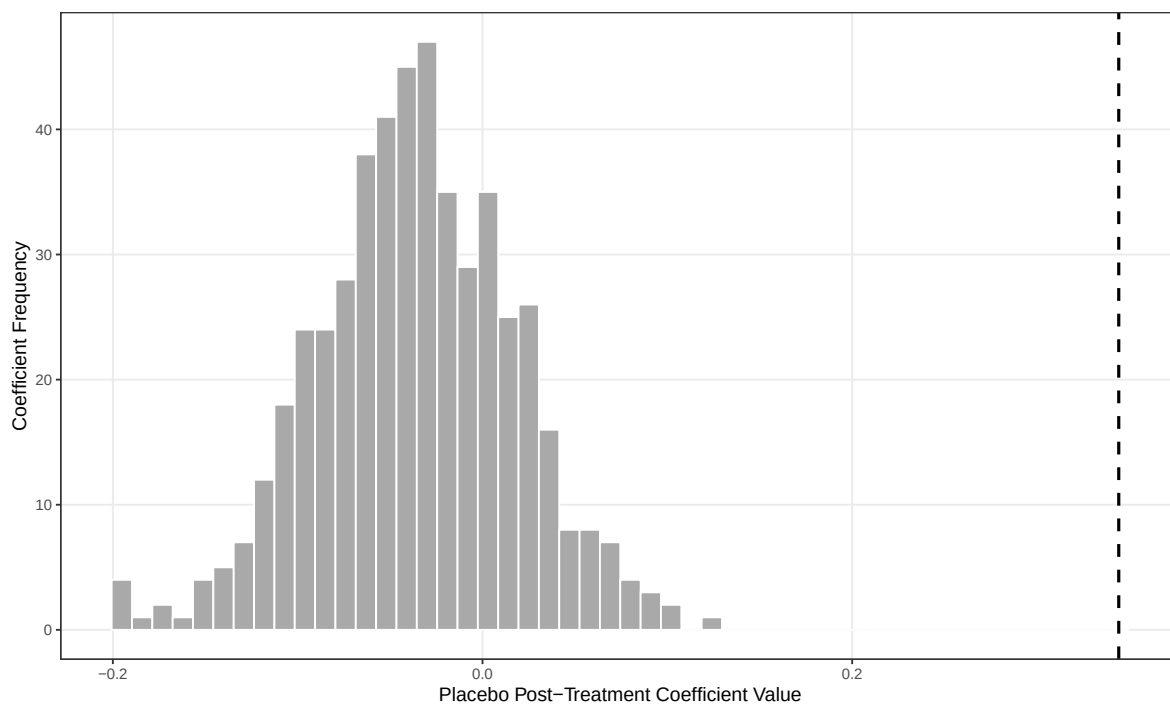


Figure 8: Distribution of Placebo Post-Intervention Estimates (Dashed Line represents Actual Estimate)